MTH 207 – Exam 2 – Fall 2026 – Dr. Kim

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Rules

- The cover page should only have your name and email. Nothing else may may be written on the cover page.
- No calculators.
- No notes, books, or other outside resources.
- No phones, smartwatches, earbuds, or any other device that communicates.
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. University mathematics focuses on the **process** more than the **answer**. For example, stating "does not exist" without giving a reason will not earn points.
- *There is no guarantee that the practice exam has exactly all the same types of problems as the actual exam. (However, the practice exam is a good study resource for the actual exam.)*

The exam length is 55 minutes. There are 100 possible points.

(8pts) 1. If $f(x) = \dfrac{x^8 + x^7 + x^6}{x^7} + \sqrt{x}$, find $f'(x)$

$f(x) = x + 1 + x^{-1} + x^{1/2}$

$f'(x) = 1 + 0 + (-1)x^{-2} + \frac{1}{2}x^{-1/2}$

parens required (order of operations)

In place of this, $\dfrac{x^7(8x^7 + 7x^6 + 6x^5) - (x^8 + x^7 + x^6)7x^6}{(x^7)^2}$ is valid

(8pts) 2. If $y = \cos(10x^2 + 6^x)$ find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = -\sin(10x^2 + 6^x) \cdot (20x + 6^x \ln 6)$

(8pts) 3. If $f(x) = x^2 \cdot \dfrac{e^x}{x + \sin x}$, find $f'(x)$

$f'(x) = x^2 \cdot \dfrac{(x + \sin x)(e^x) - (e^x)(1 + \cos x)}{(x + \sin x)^2} + \dfrac{e^x}{x + \sin x} \cdot 2x$

(8pts) 4. Starting from $y = \arcsin(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.

$x = \sin y$

$1 = \cos y \cdot \frac{dy}{dx}$

$\frac{dy}{dx} = \frac{1}{\cos y}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

$\sin^2 y + \cos^2 y = 1$

$\cos^2 y = 1 - \sin^2 y$

$\cos y = \sqrt{1-\sin^2 y}$

(8pts) 5. Starting from $y = \tan(x)$ find a formula for $\frac{dy}{dx}$, showing all work including a relevant trig identity.

$= \frac{\sin x}{\cos x}$

$\frac{dy}{dx} = \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$

(8pts) 6. If $y = \sin(\cos(8x)) + x^2 \tan x$ find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = \cos(\cos(8x)) \cdot -\sin(8x) \cdot 8 + x^2 \sec^2 x + \tan(x) \cdot 2x$

(8pts) 7. If $f(x) = \dfrac{x^6}{3x^{10}+1} + \cos(x^4)$, find $f'(x)$

$f'(x) = \dfrac{(3x^{10}+1)(6x^5) - (x^6)(30x^9)}{(3x^{10}+1)^2} + -\sin(x^4) \cdot 4x^3$

(8pts) 8. If $y = 2^{\sin x} + \sec x$ find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} = 2^{\sin x} \ln 2 \cdot \cos x + \sec x \tan x$

(8pts) 9. If y depends on x according to $x^3 + y^2 = x + y$, find $\frac{dy}{dx}$.

$$3x^2 + 2y \cdot \frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$2y \cdot \frac{dy}{dx} - \frac{dy}{dx} = 1 - 3x^2$$

$$\frac{dy}{dx}(2y - 1) = 1 - 3x^2$$

$$\frac{dy}{dx} = \frac{1 - 3x^2}{2y - 1}$$

(7pts) 10. Find an equation for the tangent line to $f(x) = 10^x$ at $x = 2$.

$f'(x) = 10^x \ln 10$

$f'(2) = 10^2 \ln 10$

$f(2) = 10^2$

This not this

$$y - 10^2 = 10^2 \ln 10 (x - 2)$$

or 100 or $100 \ln 10$

(7pts) 11. Find $\lim_{x\to 6} \dfrac{x^2-7x+6}{x-6}$ $= \lim_{x\to 6} \dfrac{(x-6)(x-1)}{(x-6)} = \lim_{x\to 6} (x-1) = 6-1 = 5$

This x-9 must be in parentheses (order of operations)

(7pts) 12. Find $\lim_{x\to 9} \dfrac{x-9}{\sqrt{x}-3} \cdot \dfrac{\sqrt{x}+3}{\sqrt{x}+3}$ $= \lim_{x\to 9} \dfrac{(x-9)(\sqrt{x}+3)}{x-9} = \lim_{x\to 9} (\sqrt{x}+3) = 3+3 = 6$

(7pts) 13. Find $\lim_{x\to 0} \dfrac{\sin(3x)}{8x} \cdot \dfrac{3}{3}$ $= \lim_{x\to 0} \dfrac{3}{8} \cdot \dfrac{\sin(3x)}{3x} = \dfrac{3}{8} \lim_{x\to 0} \dfrac{\sin(3x)}{3x}$

$\theta = 3x$

$x \to 0 \quad \theta \to 0$

$= \dfrac{3}{8} \lim_{\theta\to 0} \dfrac{\sin\theta}{\theta}$

$= \dfrac{3}{8} \cdot 1$

$= \dfrac{3}{8}$